How Far Can Public Data Reconstruct Storm-Outage Footprints?

A Connecticut validation of storm-report-based outage placement against ORNL EAGLE-I

Research brief · Abid · July 2026 · built on the CT storm-restoration simulator (with Alex Luo)

The question

Public datasets are increasingly used to model where storms knock out electric power: NOAA/NCEI records geolocated wind and tornado damage, and ORNL's EAGLE-I archive publishes county-level customers-without-power every 15 minutes. A natural question follows — **how far can public data alone reconstruct the spatial footprint of an outage event, and where does it break?** Using Connecticut as a testbed, I placed a storm's outages by proximity to real NCEI damage reports and then checked, against an independent county-resolution record, whether the outages landed in the counties that actually lost power.

Data and method

Placement (the thing being tested): outages are drawn toward real NCEI Storm Events point reports (Thunderstorm-Wind / Tornado latitude–longitude with wind magnitude), weighted by local customer exposure from U.S. Census tracts. **Ground truth (independent):** ORNL EAGLE-I county-level customers-out — the utilities' own outage-map archive, which the placement never sees. **Test:** every Connecticut storm from 2018–2024 that has both a set of NCEI point reports and a real EAGLE-I county signal (11 convective storms), scored by the Pearson correlation between the placement's per-county outage share and EAGLE-I's, across the state's 8 counties; plus Connecticut's other major outage events for context.

Finding 1 — for convective storms, public reports locate outages well

Across the 11 convective storms, report-weighted placement reproduces the real county footprint far better than a population-only baseline: **median Pearson r rises from 0.51 to 0.80**, the report-weighted method wins in **8 of 11** storms, and reaches $r \geq 0.5$ in **10 of 11**. This includes the May 2018 macroburst/tornado outbreak (135,000 customers, the third-largest event here), placed at $r = 0.96$. In plain terms: when a storm produces geolocated convective-damage reports, those public points are enough to say *which counties* lost power.

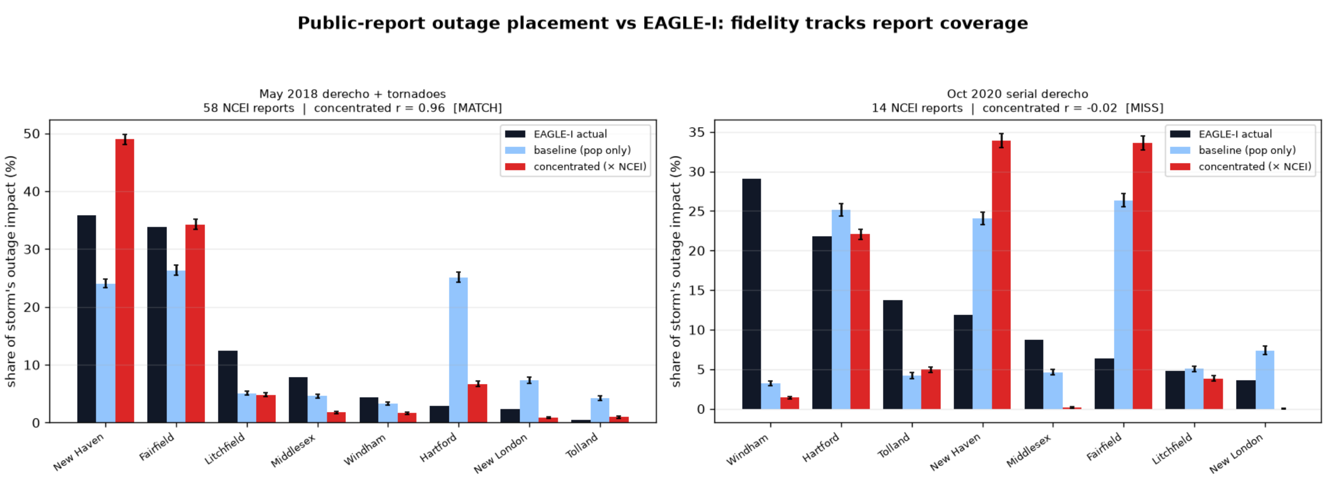

Fig. 1 — Two convective storms vs. EAGLE-I. Left: May 2018 hit the populated southwest, densely reported; report-weighted placement (red) tracks the real footprint (black). Right: the Oct 2020 derecho tracked rural NE Connecticut, which the population-biased report network under-samples — the lone failure.

Finding 2 — but public reports are blind to Connecticut's biggest storms

NCEI only geolocates *convective* damage. Tropical storms and synoptic high-wind events are logged as county-*zone* records with no coordinates; winter storms produce no wind reports at all. So for the events that cause Connecticut's largest outages, the method has essentially no data to work with:

Event	Peak customers out	Usable NCEI points
Isaias (Aug 2020) — tropical	725,700	1

Event	Peak customers out	Usable NCEI points
March 2018 nor’easters — winter	170,041	0
Halloween 2019 windstorm — synoptic	90,583	0
Oct 2019 windstorm — synoptic	45,516	0
Ida remnants (Sep 2021) — tropical	36,822	0
Henri (Aug 2021) — tropical	32,279	0

Of Connecticut’s six largest outage events in this window, three — Isaias, the March 2018 nor’easters, and the 2019 Halloween windstorm — carry near-zero usable point reports. The method is blind to exactly the storms that matter most.

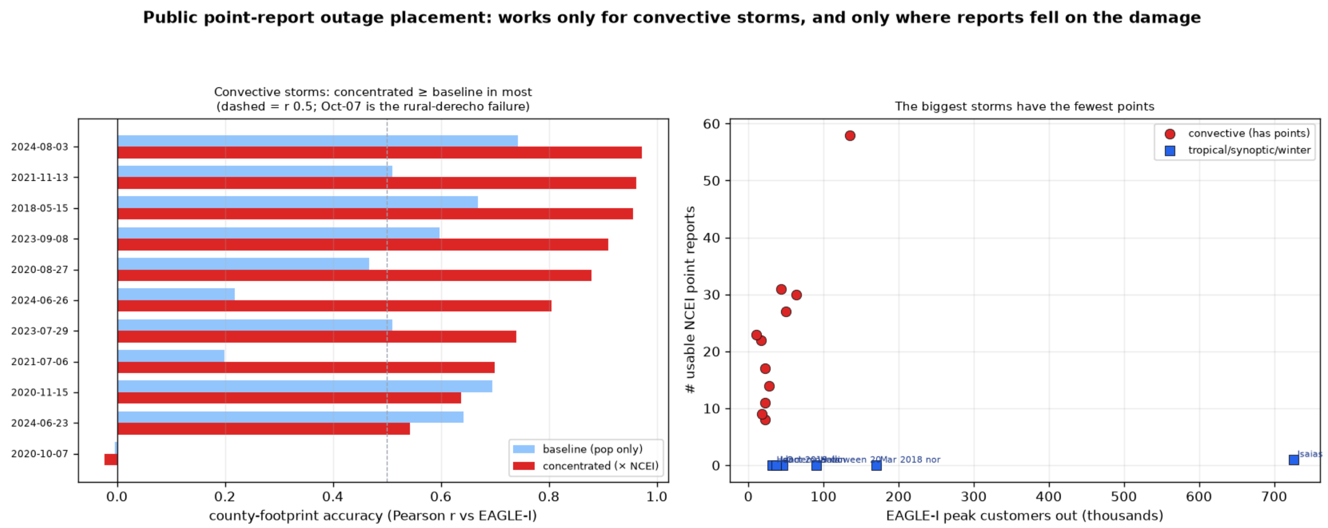

Fig. 2 — Left: per-storm county-footprint accuracy, report-weighted (red) vs. population-only (blue); dashed line r = 0.5. Right: the applicability gap — the largest events (blue) sit at zero point reports, far from the convective cluster (red).

What this means, and where I’d value your guidance

Public-report placement is a real but **narrow** tool: it reconstructs the county-level footprint of convective storms, yet is structurally blind to the tropical, synoptic, and winter events that dominate Connecticut’s outage totals, and it fails unpredictably when convection tracks rural areas the report network under-samples. This is, on real data, a concrete version of the limit EAGLE-I’s authors noted qualitatively — that reconstructing outage detail from public data alone is not generally possible. The clear way past it is data at finer-than-county resolution. A few questions where your perspective would be especially valuable:

- Is feeder- or town-level outage / restoration data (Eversource or UI) accessible for research — even for a handful of benchmark storms such as Isaias or the 2018 nor’easters?
- Would validating outage placement at *sub-county* resolution, and extending it beyond convective storms, be a contribution worth pursuing jointly?
- Which framing would you steer this toward — the “limits of public data” angle, or a modeling-improvement angle — and are there venues you’d have in mind?
- EAGLE-I is national; would adding a second utility or state materially strengthen the result?

Data & reproducibility. NOAA/NCEI Storm Events Database; ORNL EAGLE-I recorded outages 2014–2025 (doi:10.6084/m9.figshare.24237376); U.S. Census TIGER/2020 tracts. Placement and scoring code (Python, Monte-Carlo over 3,000 outages × 30 seeds per storm) available on request. Prepared as a discussion brief; figures and per-storm tables reproduce from the committed scripts.